

Abelian-square densities under bounded local avoidance: a non-monotone ternary family

Abstract

Let L_h be the ternary words containing no Abelian square of half-length $K \in \{2, \dots, h\}$; half-length 1 is permitted, so aa, bb, cc remain allowed. We ask how often a word of L_h is *itself* an Abelian square. Let $R_n^{(h)}$ be the fraction of words in $L_h \cap \Sigma^{2n}$ whose two halves have equal Parikh vector. For each $h \in \{2, \dots, 6\}$ we prove

$$R_n^{(h)} \sim C_h/n, \quad C_h = \frac{1}{2\sqrt{3}\pi a_h},$$

where a_h is the asymptotic variance rate of one letter count under the measure of maximal entropy on L_h . The proof folds the event into a rank-two additive functional of a finite Markov chain, certifies a unimodular product lattice by explicit equal-length cycles, and shows that transient window states cancel exactly between numerator and denominator.

Every C_h exceeds the free ternary value $3\sqrt{3}/(4\pi)$: within this family, bounded local avoidance raises the leading density of long Abelian squares. The dependence on h is not monotone—the effect weakens for $h = 2, 3, 4$ and grows at $h = 5, 6$. The first increase occurs at the same cutoff as the first loss of recurrent states, and a $K = 5$ ablation identifies a single half-Parikh transition class associated with both.

1 Introduction

An Abelian square is a word UV with equal-length consecutive halves satisfying

$$\Psi(U) = \Psi(V),$$

where

$$\Psi(w) = (|w|_a, |w|_b, |w|_c)$$

is the Parikh vector.

Fix the ternary alphabet

$$\Sigma = \{a, b, c\}.$$

For $h \geq 2$, define

$$L_h = \{w \in \Sigma^* : w \text{ contains no Abelian square of half-length } K \in \{2, \dots, h\}\}.$$

The half-length $K = 1$ is deliberately excluded from the forbidden set: this matches the Mäkelä formulation motivating these finite-cutoff languages, in which the trivial Abelian squares aa, bb, cc are allowed.

Our basic question has a simple form:

Among words that avoid all Abelian squares of half-length at most h , how often is the entire word itself a longer Abelian square?

For $XY \in L_h \cap \Sigma^{2n}$, $|X| = |Y| = n$, set

$$R_n^{(h)} = \frac{\#\{XY \in L_h \cap \Sigma^{2n} : \Psi(X) = \Psi(Y)\}}{|L_h \cap \Sigma^{2n}|}. \quad (1)$$

Thus $R_n^{(h)}$ is literally the density of half-length- n Abelian squares inside the bounded-avoidance language L_h .

The corresponding unrestricted problem is the natural external benchmark studied by Richmond and Shallit [7]. Independently of that attribution, the free ternary multinomial model gives the leading constant

$$C_{\text{free}} = \frac{3\sqrt{3}}{4\pi} = 0.413496\dots$$

The exact normalization against the published Richmond–Shallit [7] asymptotic is kept as a separate literature check rather than assumed here. Our main result gives the constrained constants for the first five nontrivial cutoffs:

h	C_h	C_h/C_{free}
2	0.616760	1.4916
3	0.599007	1.4486
4	0.569265	1.3767
5	0.765537	1.8514
6	0.996661	2.4103

Table 1: The certified family of leading constants.

Hence every bounded-avoidance language in the certified family has a larger leading density of long Abelian squares than the free language, but the effect is not monotone:

$$\boxed{C_2 > C_3 > C_4 < C_5 < C_6.}$$

The first increase occurs at $h = 5$, which is also the first cutoff where the canonical window presentation loses recurrent states. The timing alone is only a finite-family structural fact. The stronger evidence comes from a $K = 5$ ablation: among the three newly rejected half-Parikh classes, type $(3, 1, 1)$ is the only one incident with the lost states, deleting it alone produces the final recurrent core, and its isolated collision constant is already essentially the full L_5 value. The effects of the three classes are not additive.

The paper therefore has one main story:

$$\text{non-monotone family} \longrightarrow \text{turn at } h = 5 \longrightarrow \text{anatomy of that turn.}$$

The local-limit machinery itself is classical [1, 2, 4]. Our contribution is the bounded-Abelian-avoidance application, the explicit family of leading constants, the non-monotone cutoff dependence, and the exact finite-state structure at the first upturn.

The work is motivated by the ternary Abelian-square avoidance question associated with Mäkelä; see Rao and Rosenfeld [6] for the closest positive avoidance result. Every fully avoiding infinite word would lie in every L_h , but nothing proved here implies that such a word exists or settles the full conjecture.

Contributions

1. **Family asymptotic.** For each $h \in \{2, \dots, 6\}$,

$$R_n^{(h)} \sim C_h/n$$

with an explicit leading constant.

2. **Non-monotone cutoff dependence.** All five C_h exceed C_{free} , but the amplification weakens through $h = 4$ before increasing at $h = 5, 6$.
3. **A quantitative $K = 5$ ablation and exact core-isolation theorem.** The $(3, 1, 1)$ transition class is tied to the first recurrent-state loss and the dominant change in C ; isolating the surviving core is an exact bipartite vertex-cover problem with minimum 102.

2 Bounded-avoidance languages and window presentations

Set

$$m = 2h - 1.$$

A **valid state** is a ternary word of length m containing no Abelian square of half-length $2, \dots, h - 1$. There is an edge

$$u \longrightarrow v$$

when the two windows overlap in $m - 1$ letters and appending the new letter does not create an Abelian square of half-length h .

For every $n \geq m$, words of $L_h \cap \Sigma^n$ correspond bijectively to a choice of initial valid state together with a path of $n - m$ edges in this graph. If A_h^{full} denotes the full valid-state adjacency matrix, then

$$|L_h \cap \Sigma^n| = \mathbf{1}^T (A_h^{\text{full}})^{n-m} \mathbf{1}. \quad (2)$$

Starting from the valid graph, iteratively delete any state whose current in-degree or out-degree is zero. A state that survives is called **essential**. Equivalently, in a finite graph it can occur on a bi-infinite admissible path.

The raw, valid, and essential counts are:

h	m	raw 3^m	valid	essential	essential edges
2	3	27	27	27	66
3	5	243	162	162	360
4	7	2187	786	786	1572
5	9	19683	3114	2844	5418
6	11	177147	11070	10128	18774

Table 2: Raw, valid and essential state counts of the memory- m window presentation.

Proposition 2.1 (Certified graph hypotheses). *For every $h \in \{2, \dots, 6\}$:*

1. *the essential graph has a unique dominant strongly connected component;*

2. that component has graph period 1;
3. the projected equal-length cycle Parikh-difference lattice is all of $\mathbb{Z}^2$;
4. the associated covariance is nondegenerate;
5. the synchronized product of the time-reversed and forward core chains is primitive.

The period and lattice claims are certified by explicit closed walks. In particular, for each h , three equal-length closed walks at a common root have projected Parikh differences containing a unimodular pair. Item 5 follows because the forward and time-reversed core chains are primitive; the product of two primitive finite transition matrices is primitive as well.

The explicit roots, return lengths, cycles, Parikh vectors, and determinants are recorded in Appendix A.

Remark 2.2 (Why the non-essential part is acyclic). Every vertex on a directed cycle has in-degree and out-degree at least 1 along that cycle at every stage of the pruning. Such a vertex can therefore never be deleted. Hence every directed cycle lies in the essential set, and the non-essential part of the valid graph is necessarily a DAG.

For $h = 2, 3, 4$ there is no non-essential part. For the full L_5 presentation its longest directed path has $D_5 = 3$ edges; for L_6 , $D_6 = 5$.

3 Parry measure and the scalar variance a_h

Let A_h be the adjacency matrix of the unique dominant essential component, with Perron root λ_h and positive right and left Perron vectors r_h, ℓ_h . The measure of maximal entropy is the Parry chain

$$P_h(i, j) = \frac{(A_h)_{ij} r_h(j)}{\lambda_h r_h(i)}.$$

The S_3 -symmetry of the language forces stationary letter frequencies

$$(1/3, 1/3, 1/3).$$

Project the centered Parikh increment to two coordinates. Let Σ_h be its asymptotic covariance.

Lemma 3.1 (S_3 -equivariant covariance). *For every $h \in \{2, \dots, 6\}$,*

$$\Sigma_h = a_h \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$$

for some $a_h > 0$.

Proof. The zero-sum Parikh subspace is the irreducible two-dimensional standard representation of S_3 . The asymptotic covariance commutes with the S_3 -action and is therefore scalar on that representation. Writing the scalar operator in the coordinates (a, b) gives the stated matrix. $\square$

Consequently,

$$\det \Sigma_h = \frac{3}{4} a_h^2.$$

The values used in this paper are:

These values were obtained by a Green–Kubo/Poisson computation and independently checked by exact-moment dynamic programming. The displayed precision is intentionally shorter than the raw numerical output.

h	λ_h	a_h
2	2.4511095375	0.1489852192
3	2.2288029013	0.1534007516
4	2.0666349657	0.1614154912
5	1.9441605457	0.1200310311
6	1.8483339782	0.0921960033

Table 3: Perron roots and scalar Parikh variance rates.

4 Folding a long Abelian square

Let

$$w = w_0 w_1 \cdots w_{2n-1} \in L_h, \quad X = w_0 \cdots w_{n-1}, \quad Y = w_n \cdots w_{2n-1},$$

and let

$$Z_t = w_t \cdots w_{t+m-1}$$

be the memory- m window.

Write

$$N = n - m.$$

The fold starts at the diagonal product state

$$(Z_N, Z_N).$$

Let $\eta(u \rightarrow v) \in \mathbb{Z}^2$ be the projected Parikh vector of the new letter emitted by the forward edge $u \rightarrow v$. For $j = 0, \dots, N-1$, pair

- the left edge $Z_{N-1-j} \rightarrow Z_{N-j}$, traversed backwards, and
- the right edge $Z_{N+j} \rightarrow Z_{N+j+1}$, traversed forwards.

Define the product increment

$$g_j = \eta(Z_{N+j} \rightarrow Z_{N+j+1}) - \eta(Z_{N-1-j} \rightarrow Z_{N-j}).$$

Lemma 4.1 (Exact fold identity). *With*

$$\beta = \Psi(Z_{2n-m}) - \Psi(Z_0)$$

projected to the same two coordinates,

$$\Psi(Y) - \Psi(X) = \sum_{j=0}^{N-1} g_j + \beta.$$

Moreover,

$$\|\beta\|_\infty \leq m.$$

Proof. The paired edges account for

$$w_m, \dots, w_{n-1}$$

on the left and

$$w_n, \dots, w_{2n-m-1}$$

on the right. The only unpaired letters are the first m letters and the final m letters, whose Parikh difference is exactly

$$\Psi(Z_{2n-m}) - \Psi(Z_0).$$

Summing the paired right-minus-left emissions and the boundary term gives $\Psi(Y) - \Psi(X)$. Each state has length m , giving the uniform bound. $\square$

Lemma 4.2 (Covariance doubling). *Let $\Sigma_{D,h}$ be the asymptotic covariance of the folded difference observable. Then*

$$\Sigma_{D,h} = 2\Sigma_h.$$

Proof. Condition on the center state $Z_N = z$. Let S_N^+ be the centered projected Parikh sum to the right of the center and S_N^- the corresponding sum on the left, read backwards. Given z , the two are independent by the Markov property. The backward chain is the time reversal of the Parry chain, so it has the same asymptotic covariance:

$$\text{Cov}(S_N^\pm | z) = N\Sigma_h + O(1)$$

uniformly over the finitely many center states.

Because the observable is centered at the stationary mean and the finite primitive chain forgets its initial state geometrically fast,

$$\mathbb{E}[S_N^\pm | z] = O(1)$$

uniformly in z . The law of total covariance therefore gives

$$\text{Cov}(S_N^+ - S_N^-) = 2N\Sigma_h + O(1).$$

Divide by N and let $N \rightarrow \infty$. $\square$

The same matrix $\Sigma_{D,h}$ is the Hessian covariance appearing in the logarithm of the leading eigenvalue of the Fourier-tilted product transfer matrix introduced below; this is the standard Markov-additive covariance/pressure identification [4].

5 Endpoint-uniform LLT for the product chain

The fold evolves on the product of the time-reversed left chain and the forward right chain. Let

$$M_h(\theta)$$

denote the Fourier-tilted product transfer matrix for the rank-two increment g .

The analytic input needed here is an endpoint-uniform lattice local limit theorem. Uniformly over product-chain endpoints, bounded target displacements $v \in \mathbb{Z}^2$, and lengths $N + O(1)$,

$$\mathcal{N}_N(\alpha, \omega; v) = \mathcal{W}_N(\alpha, \omega) \frac{1}{2\pi N \sqrt{\det \Sigma_{D,h}}} (1 + o(1)), \quad (3)$$

where $\mathcal{W}_N$ is the unconstrained product-path weight/count for those endpoints. The full finite-state proof is given in Appendix F.

Product-lattice certificate

The lattice hypothesis for the product observable follows directly from Proposition 2.1. Let

$$\ell_1, \ell_2, \ell_3$$

be equal-length closed walks at a common single-chain root z_0 , with two unimodular projected Parikh differences. Pair the reversal of ℓ_i on the left with ℓ_j on the right. This is a product-chain cycle at (z_0, z_0) whose g -sum is

$$\Psi(\ell_j) - \Psi(\ell_i).$$

Choosing $i = j$ gives 0, while appropriate choices of i, j give the same unimodular pair certified in Proposition 2.1. Hence the product observable has full lattice span $\mathbb{Z}^2$.

Spectral proof scheme

For completeness, the finite-state proof uses:

1. Fourier inversion on $\mathbb{Z}^2$;
2. spectral decay of $M_h(\theta)$ away from $\theta = 0$;
3. analytic perturbation of the simple leading eigenvalue near 0;
4. zero drift from the $1/3$ stationary frequencies;
5. quadratic term $\Sigma_{D,h}$;
6. the product-lattice certificate above to rule out nontrivial peripheral characters.

This yields the displayed LLT uniformly in the finitely many endpoints and bounded displacements needed after transient peeling.

These are the standard finite-state Fourier/spectral steps; the project-specific work is the verification of primitivity, zero drift, nondegenerate covariance, full product lattice, and the endpoint-uniform bounded-displacement regime. The manuscript does not rely on a novelty claim for the LLT itself.

6 Transient cancellation and the main theorem

For this section only, write

- E_h for the dominant essential component of the full L_h presentation, and
- T_h for the non-essential states of that same L_h graph.

This notation is **not** the CORE/LOST partition used later for the L_4 scissor problem.

By Remark 2.2, T_h is a DAG. A path in the full L_h graph can meet T_h only in a bounded prefix or suffix. Indeed, a genuine

$$E_h \rightarrow T_h \rightarrow E_h$$

excursion would place the traversed T_h -vertices on a directed cycle, contradicting non-essentiality.

Lemma 6.1 (Exact full-path identity). *For every $n \geq m$, with $N = n - m$,*

$$\sum_{z,s,u,e} ((A_h^{\text{full}})^N)_{s,z} ((A_h^{\text{full}})^N)_{z,u} ((A_h^{\text{full}})^m)_{u,e} = \mathbf{1}^T (A_h^{\text{full}})^{2n-m} \mathbf{1} = |L_h \cap \Sigma^{2n}|.$$

Proof. This is pure path algebra. Splitting a length- $(2N + m)$ full-graph path after N and $2N$ edges is a bijection with the triples counted on the left. Since $2N + m = 2n - m$, the path-word bijection (2) of Section 2 gives the last equality. $\square$

Lemma 6.2 (Leading boundary factor). *For $h \in \{2, \dots, 6\}$, the leading all-language boundary multiplier in the folded LLT is*

$$B_h = 1.$$

Proof. For $h = 2, 3, 4$, $T_h = \emptyset$, so no transient peeling is needed.

For $h = 5, 6$, let D_h be the maximum path length in T_h , with

$$D_5 = 3, \quad D_6 = 5.$$

Consider one term in the exact decomposition of Lemma 6.1. By Lemma 4.1, its endpoint pair determines one required displacement,

$$v = \Psi(Z_0) - \Psi(Z_{2n-m}), \quad \|v\|_\infty \leq m.$$

Peel any transient prefix and suffix. At most D_h edges are removed at either end, so the remaining synchronized product path lies in the core, has length

$$N - O(1),$$

and its required displacement differs from v by only $O(1)$. Thus the bounded-displacement hypothesis of Section 5 applies uniformly.

Apply the endpoint-uniform product-chain LLT (3) to this peeled core term. Uniformity in endpoints, in bounded displacement, and in lengths $N + O(1)$ gives the same leading density

$$D_N = \frac{1}{2\pi N \sqrt{\det \Sigma_{D,h}}}$$

with a relative error $o(1)$ uniform over all endpoint configurations.

Thus each nonnegative term in the exact full-path decomposition equals its unconstrained path multiplicity times

$$D_N(1 + o(1)),$$

with one uniform relative error. Summing preserves the relative $o(1)$. Lemma 6.1 then gives

$$\text{numerator} = D_N |L_h \cap \Sigma^{2n}|(1 + o(1)).$$

Therefore $B_h = 1$. $\square$

The uniqueness of the dominant SCC is used here to obtain one common asymptotic density. With two dominant components carrying the same Perron root, different covariance data could prevent a single factor from being extracted uniformly.

Theorem 6.3 (Abelian-square density in the certified family). *For every*

$$h \in \{2, 3, 4, 5, 6\},$$

$$\boxed{R_n^{(h)} \sim \frac{C_h}{n}, \quad C_h = \frac{1}{4\pi\sqrt{\det \Sigma_h}} = \frac{1}{2\sqrt{3}\pi a_h}.$$

Proof. By Proposition 2.1 the synchronized product process is primitive, and the product-lattice certificate of Section 5 gives full lattice span. Lemma 4.1 folds the equal-Parikh event into a bounded-displacement rank-two additive event. Lemma 4.2 gives

$$\Sigma_{D,h} = 2\Sigma_h.$$

The endpoint-uniform product LLT at synchronized length

$$N = n - m$$

therefore contributes

$$\frac{1}{2\pi N \sqrt{\det(2\Sigma_h)}}.$$

Lemma 6.2 gives boundary factor 1. Since

$$\frac{1}{N} = \frac{1}{n}(1 + o(1)),$$

we obtain

$$R_n^{(h)} \sim \frac{1}{4\pi n \sqrt{\det \Sigma_h}}.$$

Finally $\det \Sigma_h = 3a_h^2/4$. □

Corollary 6.4 (Free-language comparison). *For the unrestricted ternary language,*

$$a_{\text{free}} = \frac{2}{9}, \quad C_{\text{free}} = \frac{3\sqrt{3}}{4\pi}.$$

Hence

$$\boxed{\frac{C_h}{C_{\text{free}}} = \frac{2/9}{a_h}}$$

for every $h \in \{2, \dots, 6\}$. In particular,

$$C_h > C_{\text{free}}$$

throughout the certified family.

Corollary 6.5 (Non-monotonicity).

$$\boxed{C_2 > C_3 > C_4 < C_5 < C_6.}$$

The first increase occurs at $h = 5$, the same cutoff at which valid and essential state counts first differ. No general causal law is claimed.

Remark 6.6 (Unrestricted benchmark). The free-language constant $C_{\text{free}} = 3\sqrt{3}/(4\pi)$ is obtained directly from the unrestricted ternary multinomial model and is reproduced by the same covariance formula with $a_{\text{free}} = 2/9$. Richmond and Shallit's [7] unrestricted Abelian-square enumeration is therefore the natural published benchmark. Matching the precise normalization of their stated asymptotic to the present R_n -notation is a literature-verification step, not an assumption used in Theorem 6.3.

7 Anatomy of the $h = 4 \rightarrow 5$ turn

We now switch to a **different graph object**: the common-memory-9 presentation of L_4 , used to ask which newly rejected $K = 5$ transitions isolate the eventual L_5 core.

To prevent confusion with Section 6, write

- CORE for the 2844-state set that survives the full $K = 5$ restriction;
- LOST = $V(L_4) \setminus \text{CORE}$ for the 270-state complement.

In the pre-deletion L_4 graph, every LOST state is both reachable from CORE and able to return to CORE. Thus CORE $\rightarrow$ LOST $\rightarrow$ CORE excursions are exactly the phenomenon studied here. This is the opposite of the transient geometry of T_h in the already-essentialized L_h graph of Section 6.

The newly rejected $K = 5$ transitions split by half-Parikh type:

type	total	CORE $\rightarrow$ CORE	CORE $\rightarrow$ LOST	LOST $\rightarrow$ CORE	LOST $\rightarrow$ LOST
(2, 2, 1)	222	222	0	0	0
(3, 1, 1)	216	72	66	66	12
(3, 2, 0)	12	12	0	0	0

Table 4: Census of the $K = 5$ rejected transitions by half-Parikh type.

Only (3, 1, 1) is incident with LOST.

$K = 5$ ablations

Let G_{320} , G_{311} , and G_{221} denote the common-memory graphs obtained by deleting only the indicated half-Parikh class.

graph	essential	edges	C
L_4	3114	6300	0.569265
G_{320}	3114	6288	0.587243
G_{311}	2844	5652	0.768856
G_{221}	3114	6078	0.514383
L_5	2844	5418	0.765537

Table 5: Single-class $K = 5$ ablations in the common memory-9 presentation.

Deleting only (3, 1, 1) produces the same essential state set as the full $K = 5$ restriction.

Define

$$\Delta_{320} = C(G_{320}) - C(L_4) = +0.0179785\dots,$$

$$\Delta_{311} = C(G_{311}) - C(L_4) = +0.1995914\dots,$$

and

$$\Delta_{221} = C(G_{221}) - C(L_4) = -0.0548822\dots$$

The full change is

$$\Delta_{\text{full}} = C(L_5) - C(L_4) = 0.1962719\dots$$

The single-class changes sum to

$$0.1626876\dots,$$

so the interaction term

$$\Delta_{\text{int}} = \Delta_{\text{full}} - (\Delta_{320} + \Delta_{311} + \Delta_{221}) = +0.0335843\dots$$

is nonzero. In particular, the $(2, 2, 1)$ class alone moves the constant in the opposite direction from the dominant $(3, 1, 1)$ change.

Also,

$$\frac{\Delta_{311}}{\Delta_{\text{full}}} = 1.016913\dots$$

This ratio is descriptive, not an additive attribution of cause. The safe conclusion is that the $(3, 1, 1)$ ablation is associated with both the entire recurrent-state loss and a collision-constant change slightly larger than the full $K = 5$ change, while the remaining deletions and their interactions partially offset it.

The induced LOST graph in the pre-deletion L_4 graph is acyclic, with 216 internal edges and longest directed path 5 edges. After the full $K = 5$ deletion, the non-essential L_5 subgraph instead has 204 internal edges and longest path 3. These are different graphs.

8 Excursion cover and the exact scissor

Let

$$E_{\text{in}} = E \cap (\text{CORE} \times \text{LOST}), \quad E_{\text{out}} = E \cap (\text{LOST} \times \text{CORE}).$$

The full boundary has

$$|E_{\text{in}}| = |E_{\text{out}}| = 180.$$

Construct a bipartite conflict graph H whose left vertices are E_{in} and right vertices are E_{out} . Join $e \in E_{\text{in}}$ to $f \in E_{\text{out}}$ whenever

$$\text{head}(e) \rightsquigarrow \text{tail}(f)$$

inside the induced LOST graph, allowing the empty path.

A vertex of H therefore represents a boundary **transition** of the original automaton.

Proposition 8.1 (Excursion-cover equivalence). *Let G be a finite directed graph, let C^* induce a strongly connected subgraph containing a directed cycle, and let $T^* = V(G) \setminus C^*$ induce an acyclic subgraph. Let E_{in} and E_{out} be **all** boundary edges between C^* and T^* , and form H as above.*

For

$$S \subseteq E_{\text{in}} \cup E_{\text{out}},$$

$$\text{Ess}(G - S) = C^* \iff S \text{ is a vertex cover of } H.$$

Proof. Suppose S covers H . If a vertex of T^* were both reachable from C^* and able to return to C^* in $G - S$, there would be a surviving entry edge e , a surviving exit edge f , and an internal T^* -path connecting them. Then e and f would form an uncovered edge of H , contradiction. Since $G[T^*]$ is acyclic, no vertex of T^* can remain essential. The subgraph induced by C^* is untouched because only boundary edges are deleted.

Conversely, if S is not a vertex cover, some adjacent entry e and exit f both survive. Their adjacency gives an internal T^* -path. Strong connectivity of C^* closes this excursion through C^* into a recurrent directed walk, making every vertex on the T^* -segment essential. Hence the essential set is strictly larger than C^* . $\square$

By König’s theorem,

$$\tau(H) = \nu(H).$$

For the $K = 5$ scissor graph,

$$|V_L(H)| = |V_R(H)| = 180, \quad |E(H)| = 366,$$

and explicit matching and cover certificates (Appendix C) give

$$\tau(H) = \nu(H) = 102.$$

Restricted (3, 1, 1) deletion pool

Only

$$66 + 66 = 132$$

of the 360 full CORE-LOST boundary transitions belong to the $K = 5$ -rejected (3, 1, 1) pool. Restricting deletions to this pool is therefore a **vertex-cover problem with forbidden vertices**, not literally the unrestricted setting of Proposition 8.1.

Nevertheless the restricted optimum remains 102. A human-readable certificate is

$$84 + 18 = 102.$$

Here 84 pool transitions are forced because their conflict partner lies outside the deletable pool. After removing them, the residual conflict graph contains 18 disjoint copies of K_2 , each requiring exactly one of its two conflict-graph vertices. Since each conflict-graph vertex represents a boundary transition, these contribute exactly 18 further deletions.

Twelve remaining pool transitions are redundant for minimum covers: they are not isolated in H , but every excursion incident with them is already broken by forced deletions.

Symmetry refinements

Under S_3 invariance the optimum is 17 transition-orbits, i.e. 102 transitions. Under S_3 plus reversal invariance the minimum is 9 reversal pairs, i.e. 108 transitions. Detailed orbit enumeration and counts of optimal solutions are given in Appendix D.

9 Numerical validation and reproducibility

Finite- n dynamic programming is used only as an independent check of the theorem-derived constants.

For $h = 2, 3, 4$, independent all-language finite- n computations and second-order extrapolations agree with the spectral constants to approximately 0.003%, 0.009%, and 0.012%, respectively.

For $h = 5$, the corrected all-language values include:

n	$nR_n^{(5)}$
20	0.6066545
30	0.6583281
40	0.6833348
50	0.6992042
60	0.7099965

Table 6: All-language finite- n control for $h = 5$.

with theorem constant

$$C_5 = 0.7655366150 \dots$$

At $n = 60$ the finite- n value 0.7099965 is approximately 7.3% below C_5 ; a second-order Richardson extrapolation of the same series lies approximately 0.13% from C_5 . Finite- n data validate, but do not prove, the asymptotic.

A previous series was accidentally core-only; the values above are the corrected all-language control.

$h = 6$ finite- n validation

The theorem-derived value

$$C_6 = 0.9966608746 \dots$$

rests on the independently certified graph, lattice, covariance, and local-limit derivation. Finite- n data are used only as an independent numerical check. The full evidence-level breakdown for $h = 6$ is given in Appendix B.

n	$ L_6 \cap \Sigma^{2n} $	collisions	$nR_n^{(6)}$
12	32,264,706	1,482,864	0.5515118594
20	598,401,288,060	19,842,060,168	0.6631690327
30	129,601,490,959,197,930	3,289,168,271,775,816	0.7613727853
40	28,068,484,992,703,808,552,340	570,703,655,838,606,639,024	0.8133016883
50	6,078,940,846,562,758,147,891,532,058	102,845,141,999,403,991,617,566,730	0.8459133309

Table 7: Verified raw all-language finite- n values for $h = 6$.

Reproducibility

Reproducibility materials comprise the graph builders, SCC and graph-period certificates, explicit single- and product-lattice witnesses, two independent covariance calculations, all-language finite- n controls, $K = 5$ ablation and cover certificates, symmetry calculations, and source hashes/commands/environments. The preserved $h = 6$ reporting defect and its arithmetic correction audit are included with these materials. Appendix E records the provenance level of each item.

10 Discussion, relation to Mäkelä, and open questions

The family gives two simultaneous facts:

$$C_h > C_{\text{free}} \quad (h = 2, \dots, 6),$$

yet

$$C_2 > C_3 > C_4 < C_5 < C_6.$$

Thus bounded local avoidance makes long Abelian squares more common than in the free language throughout the certified family, but increasing the cutoff does not increase that effect monotonically.

The identity

$$\frac{C_h}{C_{\text{free}}} = \frac{2/9}{a_h}$$

shows that the phenomenon is exactly controlled by the scalar asymptotic Parikh variance.

At $h = 5$, the first upturn occurs at the same cutoff as the first valid-to-essential state loss. The ablation goes further than this timing coincidence: the $(3, 1, 1)$ class is the only new rejection class incident with LOST, deleting it alone produces the final core, and its isolated C -shift dominates the observed change. Because the ablations are non-additive, we do not claim a general causal law.

The Perron roots decrease with h ,

$$2.4511, 2.2288, 2.0666, 1.9442, 1.8483,$$

while the collision constants do not. Entropy reduction and concentration of the Parikh distribution therefore need not move in lockstep.

Relation to Mäkelä’s problem

Every infinite ternary word avoiding all nontrivial Abelian squares would lie in every finite cutoff language L_h . Rao and Rosenfeld [6] established the closest positive avoidance result, for sufficiently large periods. The present results describe necessary finite-window containers, but they do not prove:

- existence of an infinite word surviving every cutoff;
- convergence of the finite-cutoff Parry measures;
- a positive growth rate for the full infinite-avoidance language;
- a proof or disproof of Mäkelä’s conjecture.

Related work and novelty boundary

Richmond and Shallit [7] provide the direct unrestricted Abelian-square enumeration benchmark. General symbol-count central and local limit theorems for regular languages are treated by Bertoni et al. [1] and later work [2]; multidimensional spectral Markov methods are standard [4]. Directional counting for regular languages is developed by Grigorchuk and Quint [3]. Repetition-free language literature [8, 5] supplies the surrounding growth and avoidance context.

The paper does **not** claim a new general local-limit theorem. At the present audit stage the strongest safe novelty wording remains:

We are not aware of a previous asymptotic enumeration of adjacent Abelian-square collisions inside a finite-type language defined by bounded Abelian-square avoidance.

Before submission, the citation audit should verify the exact normalization of the published Richmond–Shallit [7] asymptotic against the present R_n -notation and complete the final citation-neighborhood search.

Open questions

The present paper leaves outside its theorem package:

1. the behavior of C_h for $h \geq 7$;
2. whether later recurrent-core losses accompany sharp changes in a_h ;
3. whether analogous scissor structures recur at later cutoffs;
4. second-order terms in $R_n^{(h)}$;
5. larger alphabets and non-adjacent blocks.

No $h = 7$ computation is needed before the present manuscript is stabilized.

References

- [1] A. Bertoni, C. Choffrut, M. Goldwurm, and V. Lonati. On the number of occurrences of a symbol in words of regular languages. *Theoretical Computer Science*, 302:431–456, 2003.
- [2] M. Goldwurm, J. Lin, and M. Vignati. Local limit laws for symbol statistics in bicomponent rational models. *Theoretical Computer Science*, 970:114051, 2023.
- [3] R. Grigorchuk and J.-F. Quint. Directional counting for regular languages. *Algebra and Discrete Mathematics*, 41(1):51–109, 2026.
- [4] L. Hervé and F. Pène. The Nagaev–Guivarc’h method via the Keller–Liverani theorem. *Bulletin de la Société Mathématique de France*, 138(3):415–489, 2010.
- [5] E. A. Petrova and A. M. Shur. Branching frequency and Markov entropy of repetition-free languages. arXiv:2105.02750 [cs.FL], 2021.
- [6] M. Rao and M. Rosenfeld. Avoiding two consecutive blocks of same size and same sum over $\mathbb{Z}^2$. *SIAM Journal on Discrete Mathematics*, 2018.
- [7] L. Bruce Richmond and Jeffrey Shallit. Counting Abelian squares. *The Electronic Journal of Combinatorics*, 16(1):R72, 2009.
- [8] A. M. Shur. Growth properties of power-free languages. *Computer Science Review*, 6(5–6):187–208, 2012.

A Period and lattice certificates

This appendix records the explicit witnesses for parts 2 and 3 of Proposition 2.1. All strings below are the sequences of *emitted letters* of a closed walk read from the stated root; the walk returns to that root. Every walk was checked edge by edge against the memory- m window graph, and the state validity of the root was checked independently.

For each h the period certificate consists of two closed walks at the root whose lengths are coprime, which forces graph period 1. The lattice certificate consists of three closed walks of one common length L at the same root; writing Ψ for the Parikh vector and projecting onto the first two coordinates, the two differences $d_1 = \Psi(c_2) - \Psi(c_1)$ and $d_2 = \Psi(c_3) - \Psi(c_1)$ satisfy $|\det[d_1 \ d_2]| = 1$, so they generate $\mathbb{Z}^2$ and the projected lattice has covolume 1.

$h = 2$, $m = 3$, **root** aaa

- Period: closed walks of lengths 4 (baaa) and 5 (bcaaa); $\gcd(4, 5) = 1$.
- Lattice at $L = 6$: $c_1 = \text{bcaaaa}$, $\Psi = (4, 1, 1)$;
 $c_2 = \text{bbbbaa}$, $\Psi = (3, 3, 0)$;
 $c_3 = \text{bbcaaa}$, $\Psi = (3, 2, 1)$.
- $d_1 = (-1, 2)$, $d_2 = (-1, 1)$, $\det = 1$.

$h = 3$, $m = 5$, **root** aabaa

- Period: closed walks of lengths 6 (caabaa) and 7 (acaabaa); $\gcd(6, 7) = 1$.
- Lattice at $L = 9$: $c_1 = \text{acbcaabaa}$, $\Psi = (5, 2, 2)$;
 $c_2 = \text{acccaabaa}$, $\Psi = (5, 1, 3)$;
 $c_3 = \text{cbccaabaa}$, $\Psi = (4, 2, 3)$.
- $d_1 = (0, -1)$, $d_2 = (-1, 0)$, $\det = -1$.

$h = 4$, $m = 7$, **root** aaabaaa

- Period: closed walks of lengths 8 (caaabaaa) and 11 (cabcaaabaaa); $\gcd(8, 11) = 1$.
- Lattice at $L = 13$: $c_1 = \text{caaabcaaabaaa}$, $\Psi = (9, 2, 2)$;
 $c_2 = \text{caabccaaabaaa}$, $\Psi = (8, 2, 3)$;
 $c_3 = \text{cabbccaaabaaa}$, $\Psi = (7, 3, 3)$.
- $d_1 = (-1, 0)$, $d_2 = (-2, 1)$, $\det = -1$.

$h = 5$, $m = 9$, **root** aaabaaaca

- Period: closed walks of lengths 8 (aabaaaca) and 13 (abccaaabaaaca); $\gcd(8, 13) = 1$.
- Lattice at $L = 13$: $c_1 = \text{abccaaabaaaca}$, $\Psi = (8, 2, 3)$;
 $c_2 = \text{bbccaaabaaaca}$, $\Psi = (7, 3, 3)$;
 $c_3 = \text{bcccaabaaaca}$, $\Psi = (7, 2, 4)$.
- $d_1 = (-1, 1)$, $d_2 = (-1, 0)$, $\det = 1$.

$h = 6$, $m = 11$, **root** abbbcaaaccb

- Period: closed walks of lengths 12 (babbbcaaaccb) and 17 (acccaaabbbaaaaccb); $\gcd(12, 17) = 1$.
- Lattice at $L = 17$: $c_1 = \text{accbbabbbcaaaccb}$, $\Psi = (5, 6, 6)$;
 $c_2 = \text{baaacaabbbcaaaccb}$, $\Psi = (8, 5, 4)$;
 $c_3 = \text{bbccaaabbbcaaaccb}$, $\Psi = (6, 6, 5)$.
- $d_1 = (3, -1)$, $d_2 = (1, 0)$, $\det = 1$.

Root independence

The projected lattice generated by differences of equal-length closed walks does not depend on the root chosen. If c, c' are equal-length closed walks at z , and p is a path $z' \rightarrow z$ and q a path $z \rightarrow z'$ inside the same strongly connected component, then pcq and $pc'q$ are equal-length closed walks at z' with the same Parikh difference. A single root therefore suffices for each h .

B Finite- n validation and evidence levels

Finite- n dynamic programming validates the theorem-derived constants; it does not prove them. This appendix records exactly which quantities received which kind of check.

$h = 2, 3, 4$

Independent all-language finite- n computations, followed by second-order Richardson extrapolation of $nR_n^{(h)}$, agree with the spectral constants to approximately 0.003% ($h = 2$), 0.009% ($h = 3$) and 0.012% ($h = 4$).

$h = 5$

The all-language series is given in Table 6. It approaches C_5 slowly: at $n = 60$ the value $nR_n^{(5)} = 0.7099965$ lies approximately 7.3% below $C_5 = 0.7655366150\dots$. A second-order Richardson extrapolation of the same series lies approximately 0.13% from C_5 . We therefore do *not* claim that the finite- n series reproduces the constant directly; it is consistent with it after extrapolation.

An earlier series was accidentally computed on the essential core only. The values in Table 6 are the corrected all-language control and must not be conflated with the core-only sequence.

$h = 6$: the reporting-only defect and its evidence levels

For $h = 6$, an earlier prose table reported absolute counts that did not correspond to the preserved raw output of the collision dynamic program. A subsequent arithmetic and provenance audit established that this was a reporting-only defect, and the evidence separates into three distinct levels. First, all five denominators $|L_6 \cap \Sigma^{2n}|$ for $n = 12, 20, 30, 40, 50$ were independently reproduced by a separate exact arbitrary-precision language-count dynamic program. Second, the collision numerators for $n = 12, 20, 30$ were independently reproduced by a separate arbitrary-precision implementation. Third, for $n = 40$ and $n = 50$ the numerators were **not** recomputed by a second

independent implementation; instead the raw computation was audited for arithmetic storage safety, the largest observed typed-array cell being 2 182 364 991 314, far below the `BigUint64Array` bound $2^{64} - 1 = 18\,446\,744\,073\,709\,551\,615$, so no wraparound can have occurred. All five reported values of $nR_n^{(6)}$ are internally consistent with the preserved raw counts as exact rationals, and the ratios reported before the correction were unchanged by it. We make no claim that the $n = 40, 50$ numerators received a second independent computation.

A small- n brute-force control over $\Sigma^{\leq 13}$ and a regression of the same pipeline against the previously validated $h = 5$ all-language series both pass.

The verified raw values are reproduced in Table 7. The correction affects only the earlier prose report, not the theorem and not the raw $h = 6$ validation data.

C Excursion-cover certificates

All quantities in this appendix are produced and checked by the package-local script `verify_scissor_and_symmetry.js`, which rebuilds the memory-9 window graph from first principles and reads no external data file. Its output for the present package is reproduced verbatim in `VERIFIER_OUTPUT.txt`; the explicit matching and cover are written to `excursion_cover_certificate.json`.

Graph and boundary

quantity	value	checked
memory-9 valid states	3114	yes
L_4 edges (all overlaps)	6300	yes
CORE	2844	yes
LOST	270	yes
LOST-internal edges in L_4	216	yes
$ E_{\text{in}} $	180	yes
$ E_{\text{out}} $	180	yes
total boundary	360	yes
$ E(H) $	366	yes

Optimum, with both certificates

$$\nu(H) = 102, \quad \tau(H) = 102.$$

Two independent certificates are supplied and machine-checked:

- a **matching** of 102 pairwise vertex-disjoint edges of H , each verified to be an actual edge of H ; this gives $\tau(H) \geq \nu(H) \geq 102$;
- a **vertex cover** of 102 vertices (66 on the E_{in} side, 36 on the E_{out} side), verified to leave 0 of the 366 edges of H uncovered; this gives $\tau(H) \leq 102$.

Together the two certificates pin the optimum exactly, independently of König’s theorem. Each cover and matching vertex is stored in the JSON certificate as the pair of 9-letter windows naming the boundary transition, so the certificate can be re-checked against any independent construction of the graph.

Restricted (3, 1, 1) pool

quantity	value	checked
pool size	132	yes
H -edges with no pool endpoint	0	yes
forced (conflict partner outside the pool)	84	yes
residual components	18	yes
all residual components are K_2	true	yes
redundant pool transitions	12	yes
restricted minimum = $84 + 18$	102	yes

The zero in the second row is the feasibility check: every edge of H has at least one endpoint inside the pool, so the forbidden-vertex problem of Section 8 is solvable at all. The decomposition $132 = 84 + 36 + 12$ accounts for every pool transition.

D Symmetry and reversal orbits

The quantities below are produced and checked by the same package-local script `verify_scissor_and_symmetry.js`; the orbit representatives and the reversal pairing are written to `symmetry_orbits_certificate.json`. No external data file is read.

S_3 action on the pool

The alphabet group S_3 acts on the memory-9 window graph by permuting letters, and this action preserves the (3, 1, 1) boundary pool. The 132 pool transitions decompose into **22 orbits, every one of size 6**; both facts are checked, and closure of the action on the pool is asserted by the script (it raises an error otherwise).

Restricting deletions to unions of whole orbits gives a vertex-cover problem on the 22-vertex quotient conflict graph. Exhaustive search over all 2^{22} orbit subsets yields

$$\text{minimum} = 17 \text{ orbits} = 17 \times 6 = 102 \text{ transitions}, \quad \#\{\text{optima}\} = 8.$$

The S_3 -invariant optimum therefore attains the same value 102 as the unrestricted optimum of Appendix C; symmetry costs nothing here.

Reversal

Reversal is a directed *anti*-automorphism: an edge $u \rightarrow v$ is carried to $\text{rev}(v) \rightarrow \text{rev}(u)$. It therefore exchanges E_{in} and E_{out} , and no orbit can be fixed by it. The script checks that the induced map on orbits is an involution and that the 22 orbits form **11 reversal pairs**.

Requiring the deletion set to be invariant under S_3 and reversal means selecting whole reversal pairs. Exhaustive search over all 2^{11} selections yields

$$\text{minimum} = 9 \text{ reversal pairs} = 18 \text{ orbits} = 108 \text{ transitions}, \quad \#\{\text{optima}\} = 2.$$

What changes and what does not

constraint	unit of choice	minimum transitions	optima
none	single transition	102	—
S_3 -invariant	orbit (6 transitions)	102	8
S_3 and reversal	reversal pair (12 transitions)	108	2

Imposing reversal symmetry *raises* the deletion cost from 102 to 108 while *reducing* the number of independent choices from eight optima to two. The two effects run in opposite directions and should not be conflated.

E Reproducibility and provenance

This appendix records, for each machine-supported quantity, what artifact supports it and at what provenance level. Provenance levels are used in a deliberately restrictive sense:

PACKAGE-LOCAL, RE-EXECUTED the script ships in this package, reads no external file, and was re-run during preparation of this package; its output is included verbatim.

RECONSTRUCTED the script ships in this package, but no upstream original could be located, so byte-for-byte historical identity with the code that first produced the numbers cannot be established.

COPIED, HASH DIFFERS an upstream file of the same name was located, but its SHA-256 differs from the packaged copy; the packaged copy is therefore a variant, not a faithful archive.

INDEPENDENT RE-DERIVATION the quantity was recomputed during refereeing by a separately written implementation.

Evidence map

Boundary and symmetry certificates. Boundary counts, $|E(H)|$, $\nu(H) = \tau(H) = 102$, the restricted pool decomposition, and the S_3 /reversal optima of Appendices C and D. Artifact: `verify_scissor_and_symmetry.js`, with output in `VERIFIER_OUTPUT.txt` and explicit JSON certificates. Provenance: PACKAGE-LOCAL, RE-EXECUTED.

Period and lattice certificates. The $h = 2, \dots, 6$ witnesses of Appendix A. Artifact: the walks printed in full in Appendix A. Provenance: INDEPENDENT RE-DERIVATION; every walk was re-checked edge by edge.

Window-graph counts. Raw, valid, and essential state counts in Table 2. Artifact: `evidence/sft-container.js`. Provenance: COPIED, HASH DIFFERS.

Perron roots and variance rates. The values λ_h and a_h in Table 3. Artifact: `evidence/spectral_certifier.js`. Provenance: RECONSTRUCTED; the values were independently re-derived by two methods.

$K = 5$ ablation constants. The constants in Table 5. Artifact: `evidence/ablation_certifier.js`. Provenance: RECONSTRUCTED.

Finite- n controls for $h = 2, 3, 4, 5$. The controls summarized in Table 6 and Appendix B. Artifact: `evidence/independent_dp.js`. Provenance: RECONSTRUCTED.

Finite- n data for $h = 6$. Table 7 and the correction audit. Artifact: `evidence/exact_h6_collision_dp.js`. Provenance: COPIED, HASH DIFFERS; all five denominators were also independently re-derived.

Hash record

file	packaged SHA-256 (first 16)	upstream SHA-256 (first 16)
<code>sft-container.js</code>	def436906535ca12	6dcf2a08252eba0b
<code>exact_h6_collision_dp.js</code>	cada95990b11f3e9	bcaf8eabff09877e
<code>spectral_certifier.js</code>	dfccc7c75ebf506c	not located
<code>ablation_certifier.js</code>	3f7e8aa9530187c7	not located
<code>independent_dp.js</code>	6802da24fa551d31	not located

For the two files with a located upstream, the hashes differ; we therefore do not assert that the packaged copy is the code that originally produced the reported numbers. For the three files with no located upstream, no historical provenance claim is made at all. This is why the load-bearing quantities of Appendices A, C and D, and all five $h = 6$ denominators, were re-derived independently rather than inherited.

Reporting incident

The $h = 6$ reporting-only defect described in Appendix B is retained here rather than silently repaired: the earlier prose report quoted absolute counts that did not correspond to the preserved raw output, while the nR_n ratios were unaffected. The correction is recorded, not overwritten.

F Local limit theorem details

Throughout, $M_h(\theta)$ denotes the Fourier-tilted transfer matrix of the synchronized product chain on $E_h \times E_h$: for a product edge $E = ((x, y) \rightarrow (x', y'))$ carrying the rank-two increment $g(E) = \eta(y \rightarrow y') - \eta(x' \rightarrow x)$ of Section 4, we set $M_h(\theta)_{(x,y),(x',y')} = e^{i\langle \theta, g(E) \rangle}$ when E is an edge and 0 otherwise. Thus $M_h(0)$ is the adjacency matrix of the product graph, which is primitive by Proposition 2.1(5); write $\varrho = \rho(M_h(0))$.

F.1 Wielandt equality and the lattice obstruction

Lemma F.1. *For every $\theta \in [-\pi, \pi]^2$ with $\theta \neq 0$ we have $\rho(M_h(\theta)) < \varrho$.*

Proof. Entrywise $|M_h(\theta)| = M_h(0)$, so $\rho(M_h(\theta)) \leq \varrho$. Suppose equality holds. Since $M_h(0)$ is primitive, Wielandt’s theorem yields $\varphi \in \mathbb{R}$ and a diagonal unitary D with $M_h(\theta) = e^{i\varphi} D M_h(0) D^{-1}$;

equivalently there is $\psi : E_h \times E_h \rightarrow \mathbb{R}/2\pi\mathbb{Z}$ with

$$\langle \theta, g(E) \rangle \equiv \varphi + \psi(u') - \psi(u) \pmod{2\pi}$$

for every product edge $E = (u \rightarrow u')$. Summing along a closed product walk W of length L , the coboundary telescopes, leaving

$$\left\langle \theta, \sum_{E \in W} g(E) \right\rangle \equiv L\varphi \pmod{2\pi}. \quad (4)$$

Let c_1, c_2, c_3 be the three equal-length closed walks of length L at the common root z_0 recorded for this h in Appendix A. For each pair (i, j) let W_{ij} be the closed product walk at (z_0, z_0) that traverses c_i backwards in the first coordinate and c_j forwards in the second. Then W_{ij} has length L and, by the definition of g ,

$$\sum_{E \in W_{ij}} g(E) = \Psi(c_j) - \Psi(c_i),$$

the projected Parikh difference. Taking $i = j$ makes the left-hand side of (4) zero, whence $L\varphi \equiv 0 \pmod{2\pi}$; the constant φ is thereby eliminated with no separate argument. Substituting this back into (4) for general i, j gives

$$\langle \theta, \Psi(c_j) - \Psi(c_i) \rangle \equiv 0 \pmod{2\pi}.$$

In particular $\langle \theta, d_1 \rangle \equiv \langle \theta, d_2 \rangle \equiv 0$ for $d_1 = \Psi(c_2) - \Psi(c_1)$ and $d_2 = \Psi(c_3) - \Psi(c_1)$.

This is the only point at which the covolume of the certificate is used. Appendix A records $\det[d_1 \ d_2] = \pm 1$, so $\{d_1, d_2\}$ is a $\mathbb{Z}$ -basis of $\mathbb{Z}^2$; had the determinant been some q with $|q| > 1$, the conclusion below would only give $\theta \in (2\pi/q)\mathbb{Z}^2$ and nontrivial characters would survive. Unimodularity gives $\langle \theta, v \rangle \equiv 0 \pmod{2\pi}$ for every $v \in \mathbb{Z}^2$, hence for $v = e_1, e_2$, so $\theta \in 2\pi\mathbb{Z}^2$, i.e. $\theta = 0$ in $[-\pi, \pi)^2$ — a contradiction. $\square$

F.2 Uniform spectral decay away from $\theta = 0$

Lemma F.2. *For every $\delta > 0$ there exist $\lambda_\delta < \varrho$ and $Q < \infty$ with*

$$\|M_h(\theta)^N\| \leq Q \lambda_\delta^N \quad \text{for all } N \geq 0 \text{ and all } \theta \in K_\delta := \{\delta \leq \|\theta\|_\infty \leq \pi\}.$$

Proof. The entries of $M_h(\theta)$ are trigonometric polynomials, so $\theta \mapsto M_h(\theta)$ is continuous; in fixed finite dimension the roots of the characteristic polynomial depend continuously on the entries, so $\theta \mapsto \rho(M_h(\theta))$ is continuous. By Lemma F.1 it is strictly less than ϱ at every point of the compact set K_δ , hence

$$\lambda'_\delta := \max_{\theta \in K_\delta} \rho(M_h(\theta)) < \varrho.$$

A pointwise Gelfand estimate would not suffice here, since its implied constant depends on θ . Fix λ_δ with $\lambda'_\delta < \lambda_\delta < \varrho$ and let $C_\delta = \{z : |z| = \lambda_\delta\}$. For $(z, \theta) \in C_\delta \times K_\delta$ we have $z \notin \text{spec}(M_h(\theta))$, so the resolvent $R(z, \theta) = (zI - M_h(\theta))^{-1}$ exists; matrix inversion is continuous where defined, so $(z, \theta) \mapsto \|R(z, \theta)\|$ is continuous on the compact set $C_\delta \times K_\delta$ and

$$Q_0 := \sup_{C_\delta \times K_\delta} \|R(z, \theta)\| < \infty.$$

Since the whole spectrum of $M_h(\theta)$ lies strictly inside C_δ , the holomorphic functional calculus gives $M_h(\theta)^N = \frac{1}{2\pi i} \oint_{C_\delta} z^N R(z, \theta) dz$, whence

$$\|M_h(\theta)^N\| \leq \frac{1}{2\pi} \cdot 2\pi \lambda_\delta \cdot \lambda_\delta^N Q_0 = Q \lambda_\delta^N, \quad Q := Q_0 \lambda_\delta,$$

uniformly on K_δ . $\square$

F.3 Uniformity in endpoints, bounded displacement and lengths $N + r$

Proposition F.3. Fix $r_0 \in \mathbb{N}$ and a finite $V \subset \mathbb{Z}^2$. Let $\mathcal{N}_N(\alpha, \omega; v)$ be the number of length- N product paths from α to ω with $\sum_E g(E) = v$, and let $\mathcal{W}_N(\alpha, \omega) = (M_h(0)^N)_{\alpha, \omega}$. Then

$$\mathcal{N}_{N+r}(\alpha, \omega; v) = \mathcal{W}_{N+r}(\alpha, \omega) \frac{1}{2\pi N \sqrt{\det \Sigma_{D,h}}} (1 + o(1)),$$

where the $o(1)$ is uniform simultaneously over all state pairs (α, ω) , all $v \in V$, and all integers $|r| \leq r_0$.

Proof. Fourier inversion on $\mathbb{Z}^2$ is exact:

$$\mathcal{N}_{N+r}(\alpha, \omega; v) = \frac{1}{(2\pi)^2} \int_{[-\pi, \pi]^2} e^{-i\langle \theta, v \rangle} (M_h(\theta)^{N+r})_{\alpha, \omega} d\theta.$$

Split at $\|\theta\|_\infty = \delta$. On K_δ , Lemma F.2 bounds the integrand by $Q\lambda_\delta^{N+r}$ with $\lambda_\delta < \varrho$; this bound involves neither α, ω nor v , so that contribution is $O(\lambda_\delta^N) = o(\varrho^N/N)$ uniformly in all three parameters.

Near the origin, $M_h(0)$ is primitive, so ϱ is a simple isolated eigenvalue. By analytic perturbation theory there are $\delta_0 > 0$, an analytic simple eigenvalue $\varrho(\theta)$ and an analytic rank-one spectral projection $\Pi(\theta)$ on $\|\theta\|_\infty < \delta_0$, with the remaining spectrum inside $\{|z| \leq \varrho''\}$ for some $\varrho'' < \varrho$; hence $M_h(\theta)^{N+r} = \varrho(\theta)^{N+r} \Pi(\theta) + O((\varrho'')^{N+r})$ uniformly. Write $\varrho(\theta) = \varrho e^{p(\theta)}$ with p analytic and $p(0) = 0$. By Lemma 4.2 and the S_3 letter frequencies $(1/3, 1/3, 1/3)$ the drift vanishes, so $\nabla p(0) = 0$ and $p(\theta) = -\frac{1}{2} \theta^\top \Sigma_{D,h} \theta + O(\|\theta\|^3)$.

Substituting $\theta = \phi/\sqrt{N}$ and using $\nabla p(0) = 0$,

$$\varrho(\theta)^N = \varrho^N \exp\left(-\frac{1}{2} \phi^\top \Sigma_{D,h} \phi\right) (1 + o(1)),$$

and $\Pi(\theta) \rightarrow \Pi(0)$. Gaussian integration produces the factor $(2\pi N)^{-1} (\det \Sigma_{D,h})^{-1/2}$.

The three uniformities are now read off. *Endpoints:* there are finitely many pairs (α, ω) and the dependence enters only through the entry $\Pi(\theta)_{\alpha, \omega}$, so the maximum of finitely many $o(1)$ terms is $o(1)$. *Displacement:* v enters only through $e^{-i\langle \theta, v \rangle} = e^{-i\langle \phi, v \rangle/\sqrt{N}}$, and $|e^{-i\langle \phi, v \rangle/\sqrt{N}} - 1| \leq \|\phi\| \|v\|/\sqrt{N}$, which is uniform over the fixed finite set V . *Length:* $\varrho(\theta)^{N+r} = \varrho(\theta)^N \varrho(\theta)^r$ and $\varrho(\theta)^r \rightarrow \varrho^r$ uniformly for $|r| \leq r_0$, while $(2\pi(N+r))^{-1} = (2\pi N)^{-1} (1 + O(1/N))$. $\square$

Remark F.4. This is exactly the form consumed by Lemma 6.2. After peeling, at most D_h edges are removed at each end, so the surviving core portion has length $N - r$ with $0 \leq r \leq 2D_h$; its endpoints are the finitely many core states at which the path enters and leaves E_h ; and by Lemma 4.1 the required displacement is β adjusted by the Parikh contribution of the peeled edges, so it lies in the fixed finite set $\{v : \|v\|_\infty \leq m + 2D_h\}$. The resulting relative error is therefore a single $o(1)$ independent of the term, and since every term of the identity of Lemma 6.1 is non-negative, summation preserves it.